RAMAAKSHAY MALLIREDDY

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OBJECTIVE

Highly motivated Aerospace Engineering student with experience in automation tools, embedded/ML programming, software development, hardware manufacturing, and 3D Modeling. I am eager to apply my technical background to real-world projects and contribute to the development of innovative, scalable engineering solutions.

EDUCATION

University of Texas at Austin — **Austin, TX**
Bachelors in Aerospace Engineering — *Expected Graduation: May 2028*
Courses:

Computer Programming (Python)	Engineering Computation with MATLAB	C++ Data Structures and Algorithms
Differential Equations & Linear Algebra	AI Design and Development	Machine Learning in Robotics

WORK EXPERIENCE

IBM — **Austin, Texas**
Software/Platform Engineering Intern — *May 2026 – August 2026*

- Designed AI-powered hybrid cloud solutions for enterprise clients as part of IBM's Client Engineering team
- Implemented cloud-native workloads using Python/Java, containerization (Docker/Kubernetes), and orchestration on Red Hat OpenShift and IBM Cloud platforms
- Collaborated with cross-functional teams to translate client requirements into scalable technical architectures and proof-of-concept deliverables

Texas Spacecraft Laboratory (Electrical Power Systems Sub-Team) — *September 2025 – Current*

- Designed and developed Electrical Ground Support Equipment (EGSE) PCB integrating complex components for spacecraft power system testing and validation
- Selected and tested over 15+ power components, including batteries, solar panels, and converters, ensuring compatibility with spacecraft power requirements.
- Developed and ran orbital energy simulations predicting power consumption within ±5% accuracy across mission scenarios.

Austin Energy — **Austin, Texas**
Electrical Engineering Intern — *June 2024 – August 2024*

- Utilized AutoCAD AUD to design 10+ on-site implementations, converting customer work orders to physical designs
- Sized and designed multiple electrical components, including three-phase transformers and protective relays
- Automated over 200+ electrical grid simulations maximizing run-time efficiency for existing automation frameworks

RESEARCH EXPERIENCE

Robot Learning FRI - IL & RL Blended Recovery Architecture for Robotic Manipulation — **Austin, TX**
Robotics Research Conductor — *January 2026 – Current*

- Built a dual-policy robotic control framework in PyTorch, combining a behavior cloning (IL) policy with a deep RL recovery policy (actor–critic) to handle out-of-distribution states in dynamic environments.
- Implemented safety-constrained action projection (shielding layer) to bound control inputs and prevent collisions.
- Evaluated performance in simulated manipulation tasks, measuring collision rate, recovery latency, and task success under novel obstacle configurations.

National High School Journal of Science — **Austin, TX**
Scientific Research Conductor — *May 2025 – August 2025*

- Conducted research, focusing on optimal methods of celestial body detection
- Designed experiments testing **10+** different methods of celestial body detection: telemetry, astrometry, radio wave detection, etc.
- Published research in **a peer-reviewed journal**, which has been cited and utilized as a data source in subsequent scientific experiments.

CLUB EXPERIENCE

AIAA Aircraft Design Team — *August 2025 - Current*
Propulsion Engineer

- Selected and integrated 2 high-thrust jet engines to achieve >250 N total thrust, enabling catapult-assisted launch and sustained Mach 1.6 dash performance for the DC25 aircraft.
- Modeled and optimized fuel and thermal management systems using MATLAB and ANSYS, reducing avionics operating temperature by 18% and increasing system endurance by 22%.
- Refined engine nacelle geometry and intake placement, improving thrust-to-weight ratio by 15%.

SKILLS

Programming and Software: Python (PyTorch), C, C++, MATLAB, Java, GIT, Linux, SolidWorks, SQL, HTML
Other: Oscilloscope, waveform generator, soldering, Microsoft Office, academic research writing, presentation skills